

My Notebook
Gamma spectroscopy/1

PDF Version generated by
Ariel Smyrichinsky (arismyri@iu.edu)
on
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Ariel Smyrichinsky (arismyri@iu.edu) - Sep 07, 2026, 6:44 PM EDT

08.25.2025

1:46 PM: Garrett Willis, Camden Bush, **Ariel Smyrichinsky**

The goal of this lab is to become more familiar with particle detectors and how high energy photons/gamma rays interact with them; this will be done with a photomultiplier tube (PMT), scintillator, Nuclear Instrumentation Modules, an oscilloscope, radioactive sources, lead shielding, and Python for data analysis.

The procedure for the lab will be as follows:

1. Start with the PMT (whose make is unknown, because it is covered in metal and electrical tape; it has a red "IU MODERN PHYSICS" sticker on it) being used for this lab is already coupled to a thallium-doped sodium iodide crystal that is 3 inches in diameter.
2. Verify polarity and that the PMT/scintillator combination is wired correctly to the high voltage supply and Nuclear Instrumentation Modules (NIMs). The NIMs should also be wired to an oscilloscope and a computer with MAESTRO, a spectra analysis software.
3. Power on everything.
4. Take an energy spectrum with no source in proximity of the detector to determine background.
5. Obtain a Cs-137 source, take an energy spectrum with it using the MAESTRO software, and be able to identify notable characteristics of the spectrum: photo peaks, Compton edges, how amplifier settings affect the final recorded spectrum, altering the distance of the source from the detector.
6. Repeat the same process for Na-22 and Co-60, then analyze the results using the provided Python code, with the goal of having a calibrated energy scale determined by the energy spectra taken.

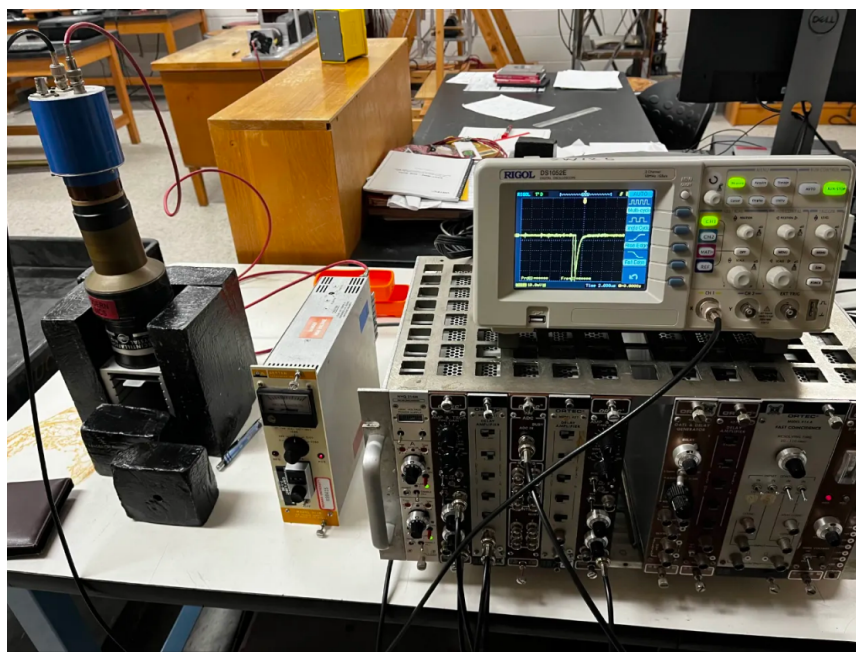
Ariel Smyrichinsky (arismyri@iu.edu) - Sep 08, 2026, 8:41 AM EDT

2:15 PM

Willis has booted up the computer with the MAESTRO software, and the high voltage supply (Bertan Associates Inc. Model 313B) which is connected to the PMT/scintillator combination. The output from the PMT goes to the amplifier (ORTEC 590A), which goes to the Analog to Digital Converter (ADC) (ORTEC 928), which has an USB output, and that goes to the USB input.

PMT voltage: 1100 volts

Amplifier gain: 51.5



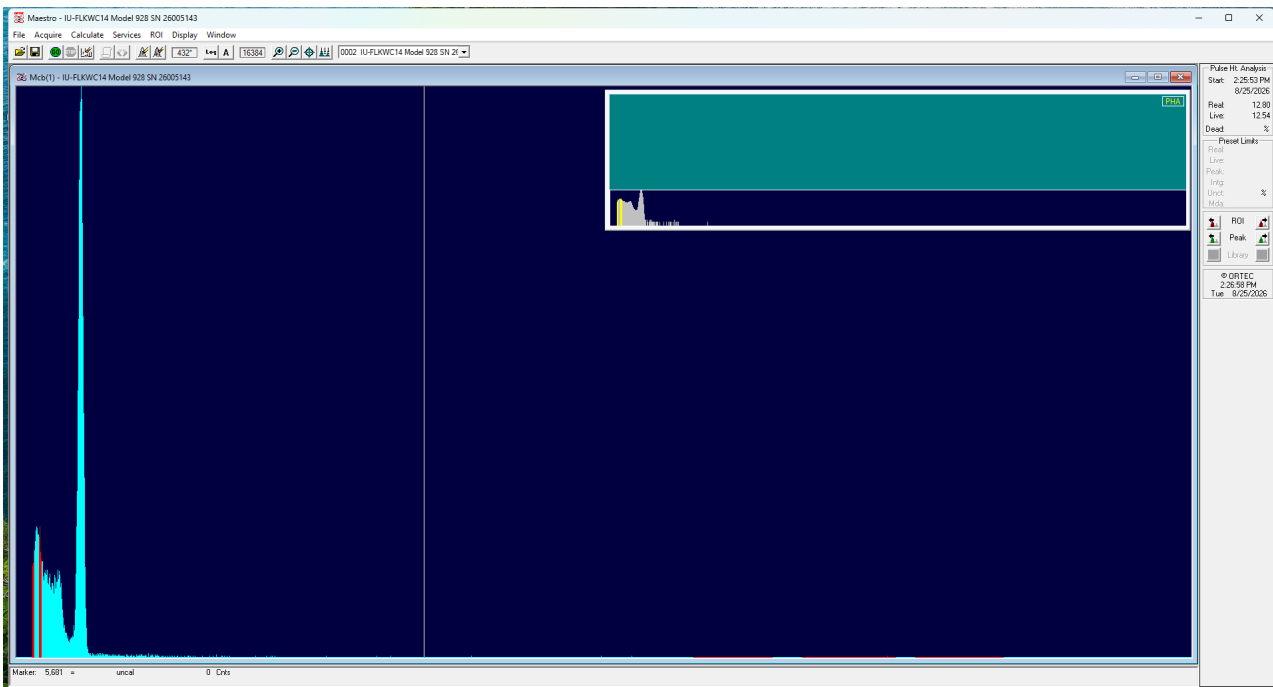
The output from the PMT can be connected directly to an oscilloscope (RIGOL DS1052E). Since there is currently no source within near proximity of the detector, the only signals detected are cosmic rays.

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2:25 PM

We are being given the Cs-137 source. We are putting it in the source shelf closest to the scintillator face.

We have hit [run] on MAESTRO and taken a preliminary spectra.



Now, we are experimenting with the amplifier, and watching the output on the oscilloscope.

Fine gain: raising it and lowering it makes small changes in the signal shape. Lower gain means less signals are detected.

Course gain: Signal height(strength) on the oscilloscope grows.

Adjusting window and lower level: no discernable change?

Ariel Smyrichinsky (arismyri@iu.edu) - Sep 07, 2026, 6:52 PM EDT

3:06 PM

As seen in the above screen capture of the Cs-137 spectra, the spectra is condensed in the left side.

We want to adjust the amplifier so that the spectra appears more distributed in the MAESTRO software. We are going to start by adjusting the fine gain, and then the coarse gain.

Initially, adjusting fine gain, and running the software for five seconds +/- 0.5 s. (determined using the built in MAESTRO timer) to discern the spectra reaction to the change in gain. Starting at 51.5. Right now, Bush is manning the amplifier and Willis is capturing the spectra.

Gain adjusted to 52.5. The spectra looks slightly stretched out.

Gain adjusted to 100.5. The spectra appears more squashed together, more so than the image.

Gain adjusted to 20.5. No spectra appears.

Gain adjusted to 200.5. Looks similar to 100.5 spectra.

Returning gain to 51.5. Now, we will adjust the voltage the high-voltage supply provides (initially 1100 V).

Adjusting voltage to 1000 V. The spectra looks even more squished, and it's getting less total counts. This is expected, because a lower voltage means that less gammas will be detected.

Adjusting voltage to 1200 V. The spectra is easier to distinguish; it's more stretched.

Adjusting voltage to 1300 V. The spectra is even easier to distinguish.

Adjusting voltage to 1350 V. The spectra is a slight improvement from the 1300 V.

Took data for an extended period of time. MAESTRO lists two time values, real and live.

Real: 62.68 seconds Live: 60.68 seconds.

Ariel Smyrichinsky (arismyri@iu.edu) - Sep 07, 2026, 6:34 PM EDT

3:46 PM

We chose to take one energy spectra per data source so we can analyze it before the next lab session. Right now, we are taking an energy spectrum of the Cs-137 source with it set on the tray on the level holder as close to the detector as possible. We are taking data for one minute, by watching the "Live time:" value on the MAESTRO user interface.

Willis is exporting the data files from the lab computer and will send them to us via email.

This spectra's name is "08_25_2026 Cs 137 1 min plot GDW.Txt" (where GDW is Willis' initials.)

3:49 PM

We have switched it out with the Na-22 source.

File name: "08_25_2026 N 22 1 min plot GDW.Txt"

3:55 PM

We have switched it out with the Co-60 source.

File name: "08_25_2026 Co 60 1 min plot GDW.Txt"

4:00 PM

We have switched it out with the unknown source.

File name: "08_25_2026 Unknown source 1 min plot GDW.Txt"

Stopping here. Shutting off the lab computer, power supply, and NIM crates.

Ariel Smyrichinsky (arismyri@iu.edu) - Sep 07, 2026, 7:09 PM EDT

08.25.2026

8:43 PM **Ariel Smyrichinsky**

We were provided an analysis notebook in Python. I am working with this in Visual Studio code as a Jupyter notebook.

I have downloaded the notebook from the lab class Canvas and changed the name.

P451_B_Gamma_Spec.ipynb has been renamed to smyrichinsky_gamma_spec_p451.ipynb.

After looking through it, I have gone through and changed some variable names in the analysis, as well as made some comments easier to follow.

I now understand why the data file looks the way it does.

The first three lines of the file are a blank line, a descriptive header, and a metadata line.

Spectrum data begins on the fourth line. Each line of data has eight values.

The first value corresponds to the first bin in that respective line, and the other values are the bin counts.

The first bin is called bin 0.

I also downloaded the spectra data that Willis shared via email, and changed the file names:

08_25_2026 Cs 137 1 min plot GDW.Txt has been renamed to smyrichinsky_cs_137_08_25_2026.txt

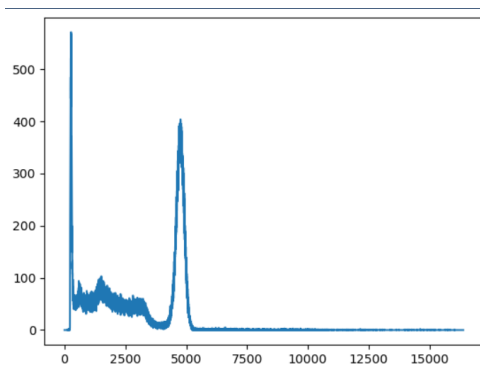
08_25_2026 N 22 1 min plot GDW.Txt has been renamed to smyrichinsky_na_22_08_25_2026.txt

08_25_2026 Co 60 1 min plot GDW.Txt has been renamed to smyrichinsky_co_60_08_25_2026.txt

08_25_2026 Unknown source 1 min plot GDW.Txt has been renamed to smyrichinsky_unknown_08_25_2026.txt

The data is read in with the already defined read_maestro_text function.

Using another pre-defined function named test_read_maestro_text, I was able to graph a preliminary Cs-137 spectra.



Ariel Smyrichinsky (arismyri@iu.edu) - Sep 07, 2026, 8:04 PM EDT

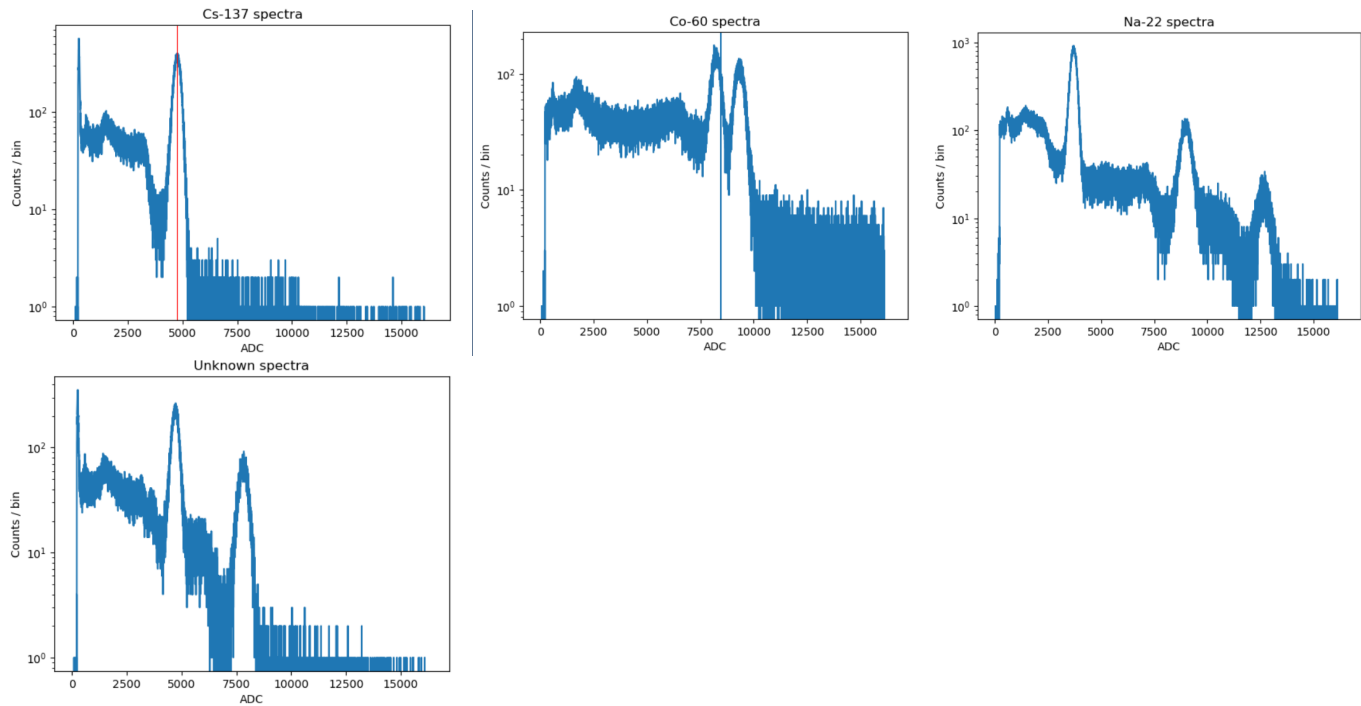
09.01.2026

1:15 PM Camden Bush, Garrett Willis, **Ariel Smyrichinsky**

Working on getting my iteration of the analysis notebook to be organized nicely.

1:47 PM

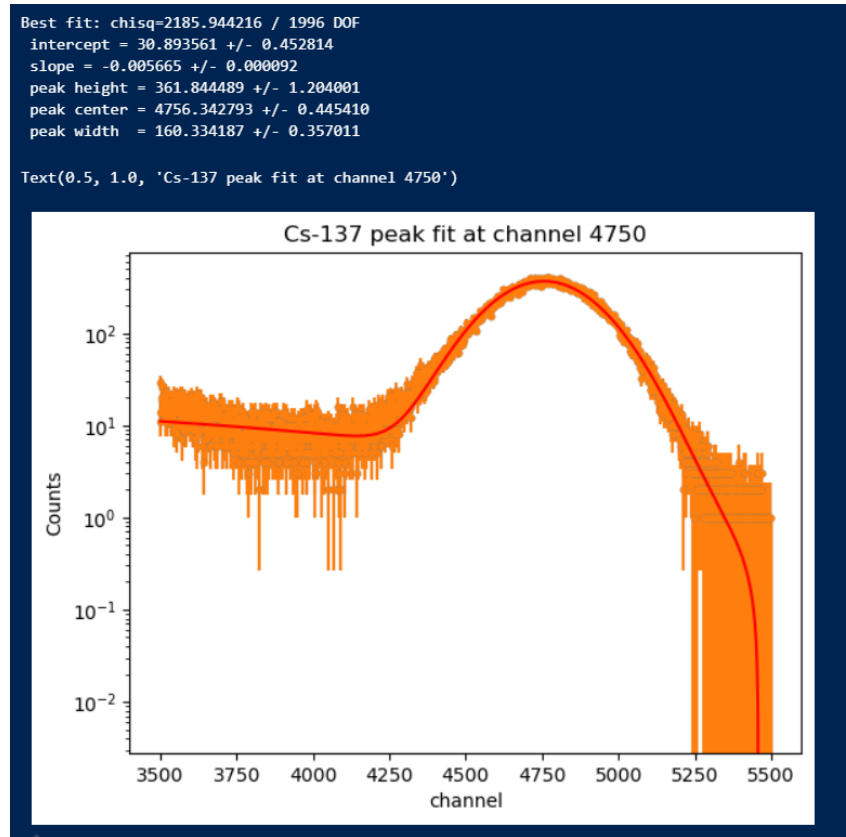
I have graphed the spectrums using the data we took last week.



The graphed vertical lines for the Cs-137 and the Co-60 spectrum should be disregarded, as I was plotting them to try and visually identify the peaks, but that is not necessary, as the pre-written fitting functions will be used for that.

I have the notebook organized such that each source has its own tab. Under each source's respective tab, I am doing the fitting for the most prominent peaks.

For fitting the Cs-137 peak that occurs at around channel 4750, I used the pre-written fit_peak function, which fits a gaussian function with a linear background to the data. According to <http://www.lnhb.fr/Laraweb/>, this is the 661 keV gamma peak because of the high recorded intensity. I used a fitting window of 3500-5500.



¹³⁷Cs - Decay data

Data Emissions Tools Radioactive chain Scheme

Data

Element: Cesium (Z=55)
Daughter(s): Ba-137 (β⁻, 100%)
Q^{*}: 1175.63 keV
Possible parent(s): Xe-137 (β⁻, 100%)
Half-life (T_{1/2}): 30.018 (22) a ≡ 947.3 (7) 10⁶ s
Decay constant (λ): 731.7 (5) 10⁻¹² s⁻¹
Specific activity (A_m): 3.2165 (24) 10¹² Bq.g⁻¹
Reference: CEA/LNE-LNHB - 2023
Citation: X. Mougeot *et al.*, Metrologia 62 (2025) 029002
Associated data files: [Table](#) - [Comments](#) - [ENSDF](#) - [PenNuc](#) - [BetaShape](#)
Data and emissions file (ASCII text format): [Cs-137.lara.txt](#)

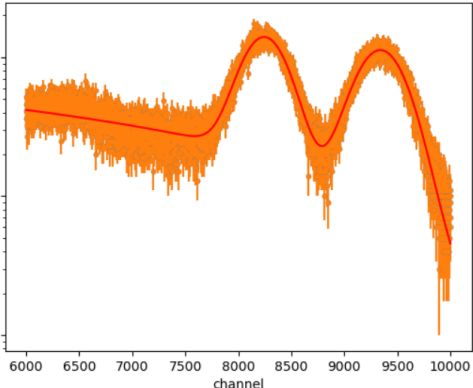
Emissions

Emissions (8 lines) sorted by increasing energy

Energy (keV)	Intensity (%)	Type	Origin [*]	Levels Start [*] End [*]
4.8824 (-)	0.919 (16)	X _L	Ba	
31.8174 (-)	1.99 (4)	X _{Kα2}	Ba	
32.1939 (-)	3.66 (6)	X _{Kα1}	Ba	
36.4457 (-)	1.078 (20)	X _{Kβ1}	Ba	
37.3317 (-)	0.272 (8)	X _{Kβ2}	Ba	
283.46 (7)	0.0006 (1)	γ	Ba-137	1 0
378.20 (7)	0.0000106 (9)	γ	Ba-137	2 1
661.6553 (30)	85.01 (20)	γ	Ba-137	2 0

```
Best fit: chisq=4446.931795 / 3996 DOF
intercept = 99.651075 +/- 0.859547
slope = -0.009648 +/- 0.000116
peak1 height = 119.909933 +/- 0.705434
peak1 center = 8246.235502 +/- 1.103874
peak1 width = 202.499597 +/- 1.054126
peak2 height = 103.091604 +/- 0.596721
peak2 center = 9343.887198 +/- 1.187251
peak2 width = 223.731957 +/- 1.450473
```

Text(0.5, 1.0, 'Co-60 peak fits in channels 8000-9500')



60Co - Decay data									
Data	Emissions	Tools	Radioactive chain	Scheme					
Data									
Element: Cobalt (Z=27)									
Daughter(s): Ni-60 (β ⁻ , 100%)									
Q ⁻ : 2823.07 keV									
Possible parent(s): Co-60m (I.T., 99.75%)									
Half-life (T _{1/2}): 5.2711 (8) a ≡ 166.340 (25) 10⁶ s									
Decay constant (λ): 4.1671 (6) 10⁻⁹ s⁻¹									
Specific activity (A _m): 41.824 (6) 10¹² Bq.g⁻¹									
Reference: INEEL - 2006									
Citation: Monographie BIPM-5, Vol. 3 (2006)									
Associated data files: BIBL - Comments - ENSDF - PenNuc - BetaShape									
Data and emissions file (ASCII text format): Co-60.lara.txt									
Emissions									
Emissions (10 lines) sorted by increasing energy									
Energy (keV)	Intensity (%)	Type	Origin [*]	Levels					
				Start [*]	End [*]				
0.84 (-)	0.0002 (-)	X _L	Ni						
7.46097 (-)	0.00334 (12)	X _{Kα2}	Ni						
7.47824 (-)	0.0065 (3)	X _{Kα1}	Ni						
8.2967 (-)	0.00136 (5)	X _{Kβ1}	Ni						
347.14 (7)	0.0075 (4)	Y	Ni-60	3	2				
826.10 (3)	0.0076 (8)	Y	Ni-60	2	1				
1.173.228.(3)	99.85 (3)	Y	Ni-60	3	1				
1.332.492.(4)	99.9826 (6)	Y	Ni-60	1	0				
2 158.57 (3)	0.0012 (2)	Y	Ni-60	2	0				
2.505.692.(5)	0.0000020 (4)	Y	Ni-60	3	0				

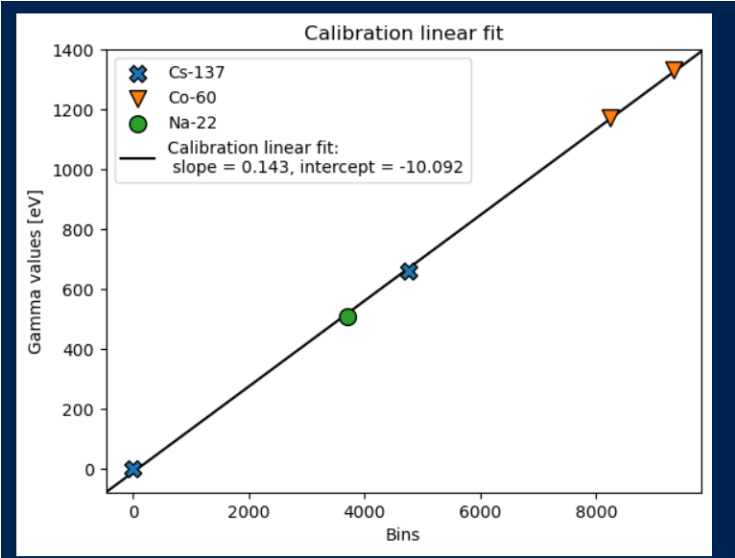
Best fit: $\text{chisq}=3770.007339$ / 2496 DOF
intercept = 80.649444 +/- 0.706741
slope = -0.012063 +/- 0.000173
peak height = 814.477766 +/- 2.008472
peak center = 3704.267952 +/- 0.294112
peak width = 138.218217 +/- 0.246255

Text(0.5, 1.0, 'Na-22 peak fit in channels 3500-4000')

Na-22 peak fit in channels 3500-4000

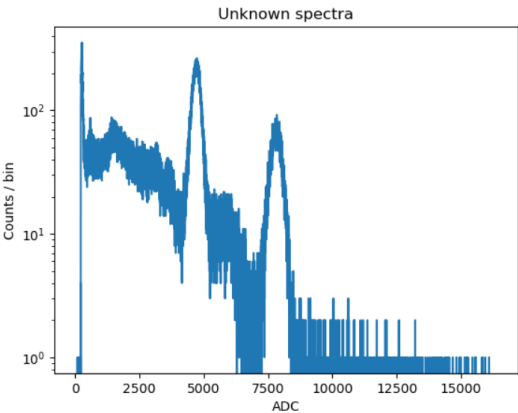
22Na - Decay data

To be able to get the detector calibration, I graphed the bin values for the fits vs the actual keV values the peaks occur at, to get a scaling factor of 0.143 and offset of -10.092.

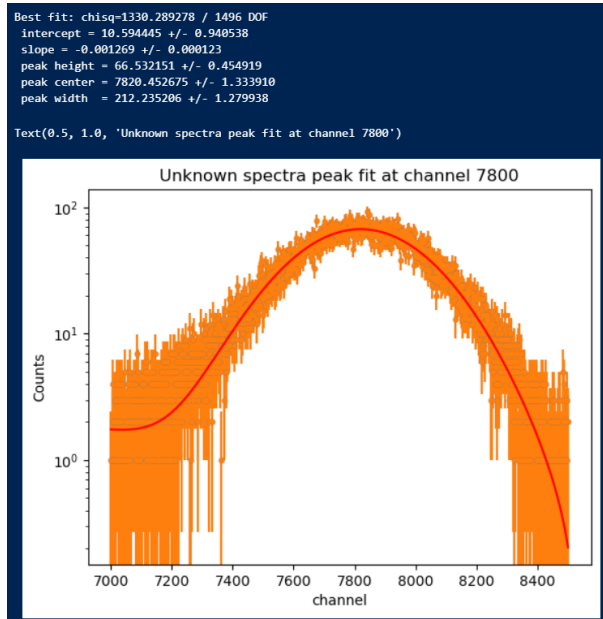
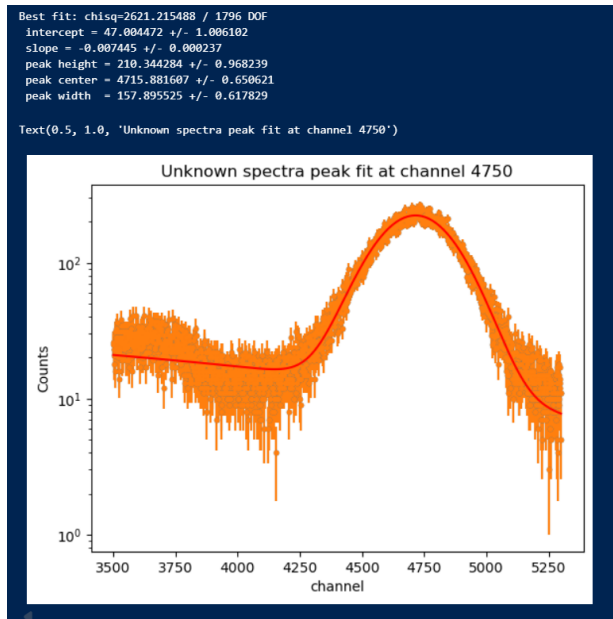


Ariel Smyrichinsky (arismyri@iu.edu) - Sep 07, 2026, 8:23 PM EDT

Next, to determine the nuclide of the unknown spectrum.



I performed fits on the two most prominent peaks that occur at around channels 5000 and 7500 respectively using the `fit_peak` function. I used fitting windows of 3500-5300 and 7000-8500 respectively.



To use the conversion values from above, the following calculation has to be done:

[Value in keV] = [scaling factor][peak bin location] + [offset]

Calculation for the peak at 4715 keV:

[Value in keV] = $(0.143)(4715 \pm 0.65) + (-10.092) = 664.153 \pm 10.09$ keV

Calculation for the peak at 7820 keV:

[Value in keV] = $(0.143)(7820 \pm 0.45) + (-10.092) = 1108.168 \pm 9.99$ keV

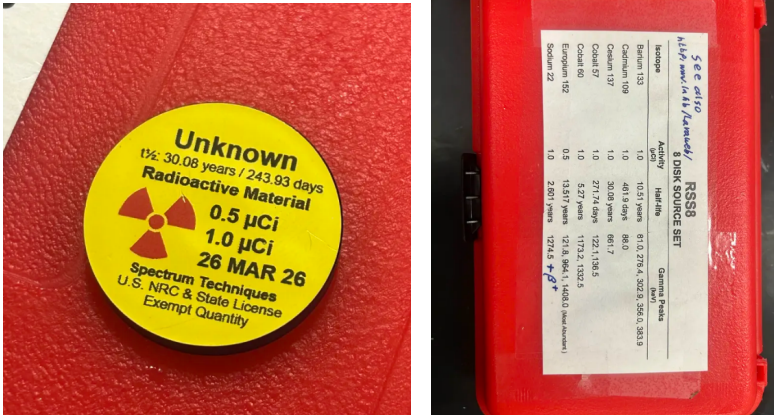
Ariel Smyrichinsky (arismyri@iu.edu) - Sep 07, 2026, 8:27 PM EDT

09.07.2026

8:08 PM, **Ariel Smyrichinsky**

When plugging the above unknown spectra values in keV at the same time into the <http://www.lnhb.fr/Laraweb/> search, using the AND function, I got no nuclide results that made sense. Then, when discussing with Willis, he suggested it could be two different sources. When searching the unknown spectrum keV values individually, we think the 664.153 keV value could be the 661 keV peak of Cs-137, which fits within uncertainty. But what about the 1108.168 keV value?

We inspected the source to see that it lists two different half lives.



The first half life of 30.08 years matches with the unknown source containing Cs-137. But the second half life of 243.93 days matches with Zn-65, which has a gamma of 1115 keV, which matches with the above value within uncertainty.

⁶⁵Zn - Decay data

DataEmissionsToolsRadioactive chainScheme

Data

Element: Zinc (Z=30)
Daughter(s): Cu-65 (β⁺, e⁻, 100%)
Q⁺: 1352.1 keV
Half-life (T_{1/2}): 244.01 (9) d ≅ 21.082 (8) 10⁶ s
Decay constant (λ): 32.878 (12) 10⁻⁹ s⁻¹
Specific activity (A_m): 304.61 (11) 10¹² Bq.g⁻¹
Reference: CEA/LNE-LNHB, INEEL - 2005
Citation: Monographie BIPM-5, Vol. 3 (2006)
Associated data files: [Table](#) - [Comments](#) - [ENSDF](#) - [PenNuc](#) - [BetaShape](#)
Data and emissions file (ASCII text format): [Zn-65.lara.txt](#)

Emissions

Emissions (8 lines) sorted by increasing energy
[Emissions meeting search criteria](#) [Other significantly intense emissions](#)

Energy (keV)	Intensity (%)	Type	Origin*	Levels Start*End*
0.9165 (-)	1.305 (21)	X _L	Cu	
8.02792 (-)	11.76 (13)	X _{Kα2}	Cu	
8.04787 (-)	22.91 (22)	X _{Kα1}	Cu	
8.9412 (-)	4.82 (7)	X _{Kβ1}	Cu	
344.95 (20)	0.00254 (18)	γ	Cu-65	21
511 (-)	2.842 (13)	γ [±]	Cu-65	-1-1
770.64 (9)	0.00269 (22)	γ	Cu-65	10
1 115.539 (2)	50.22 (11)	γ	Cu-65	20

Ariel Smyrichinsky (arismyri@iu.edu) - Sep 07, 2026, 10:13 PM EDT

Conclusion

In this lab, our group familiarized ourselves with detectors, NIM crates, radioactive sources, and using fitting procedures in order to analyze gamma energy spectra. We started with a Cs-137 source, and varied the distance from the source to the detector and noted that increasing the distance lessens the intensity of the detected gammas. To investigate how changing the amplifier affects the signal from the PMT, we connected the PMT output to the oscilloscope so we could see the changes more directly. We found that raising and lowering the fine gain made small changes in the signal shape; lower gain makes it such that the signals are amplified less, and raising the gain raises the amplification. The coarse gain does something similar, but with a stronger effect. Again, raising the gain raises the amplification.

Then, we used the MAESTRO software to study the Cs-137, Na-22, and Co-60 spectrums, as well as the spectrum of an unknown source. Once the spectrums were exported from MAESTRO, we used a pre-written Jupyter Notebook to

analyze them. The goal in our analysis was to apply fits to the prominent photopeaks in the spectrums, discern what gammas cause the photopeaks using a provided database, find the scaling factor for the detector, and then determine what the unknown source consists of.

After applying the fit functions and confidently learning the location of the peaks, they could be searched for in the database to discern which peaks were which. The procedure for this was using process of elimination for intensity. The intensities of some gammas were considerably higher than others, which made it obvious if it would be prominent in the spectra or not.

Then, in order to get the detector calibration, I graphed the bin values for the fits vs the actual keV values the peaks occur at, to get a scaling factor of 0.143 and offset of -10.092. These values were used to determine the energy values of the prominent peaks in the unknown spectrum, and we found that the spectrum had prominent peaks of 664.153 keV, which matches within uncertainty to Cs-137, and 1108.168 keV value, which matches within uncertainty to Zn-65, which has a gamma of 1115 keV.